Aayush Rai

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Education

Northeastern University Expected May 2028
Master of Science in Computer Science *Boston, MA*

Kansas State University August 2026
Bachelor of Science in Computer Science and Math Dual Major | GPA: 3.6 *Manhattan, KS*

- **Relevant Coursework:** Web Development, Data Structures and Algorithms, Object Oriented Programming, Prob & Stat, Database Systems, Operating Systems, Computer Architecture, Software Design.

Skills

- **Languages:** Python, Java, C#, SQL, JavaScript, TypeScript, HTML/CSS
- **Frameworks:** React.js, Node.js, FastAPI, ASP.NET Core
- **ML/AI:** PyTorch, TensorFlow, Scikit-learn, OpenCV, YOLOv11, LangChain, ChromaDB
- **Tools:** Git, Linux, REST APIs, SQL Server, Pandas, NumPy

Experience

Website Team Developer | *Hack for LA* July 2026 – Present

- Audited profile data for 30+ civic-tech teams on hackforla.org and merged 2 PRs correcting leadership rosters and tech stacks, resulting in accurate listings for the site's 1,000+ monthly visitors.
- Reviewed contributor PRs for YAML, formatting, and branching errors so that the merge team could approve faster.

Machine Learning Researcher | *ARISE, Kansas NSF* Nov 2025 – July 2026

- Engineered a RAG pipeline using LangChain, ChromaDB and transformer-based embedding to semantically index and query large-scale emergency response documents achieving 90%+ retrieval precision.
- Architected an interface using Gradio and FastAPI with asynchronous Python orchestration to handle concurrent stakeholder queries, maintaining less than 10 sec response latency across REST API calls on a Linux deployment environment.

Computer Vision Researcher | *CNAP, Dept. of Computer Science* Feb 2025 – May 2025

- Developed and trained computer vision pipelines using YOLOv11, TensorFlow and DeepLabCut on high-volume datasets to improve pose estimation accuracy resulting in higher mean Average Precision across validation datasets.
- Automated end-to-end data preprocessing using Python, OpenCV, Pandas, NumPy and JSON to CSV transformation pipelines so that raw experimental data could be standardized resulting in a 40% acceleration in dataset labeling.

Projects

LLMGuard | *AI Model Attestation* Spring 26

- Formulated an attestation framework with TPM and IMA file integrity with TEE-backed behavioral attestation for runtime memory integrity, defending locally deployed transformer LLMs against 4 distinct classes of integrity attacks on Linux.
- Modeled the end-to-end threat surface demonstrating that complementary defense layers are required to mitigate identified attack vectors spanning disk-level tampering and microarchitectural memory exploits.

BookFlix | *Full Stack Digital Library System* Fall 25

- Architected a normalized SQL Server schema with indexing and ACID-compliant transactions to manage 2,000+ records, ensuring data integrity for concurrent user reviews and authentication.
- Developed backend services in ASP.NET Core (C#) using direct SQL execution to ensure reproducible query behavior and integrated a React, TypeScript frontend enabling search, reviews, user authentication, and dynamic book discovery.

AguaCrop | *Kansas State TAPS Hackathon 1st Place* Fall 24

- Designed and implemented a hyper-trained RAG-based chatbot using Python and natural language processing techniques to deliver interactive, real-time irrigation insights leveraging Node.js for backend API and server-side data processing.
- Built predictive models using Scikit-learn on historical agricultural and weather data and integrated output with a React.js frontend via REST APIs to visualize irrigation schedules, delivering a functional MVP within a 48-hour sprint.